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Web Browser Navigation Using Stack (ADT)

CSA0305 – DATA STRUCTURES

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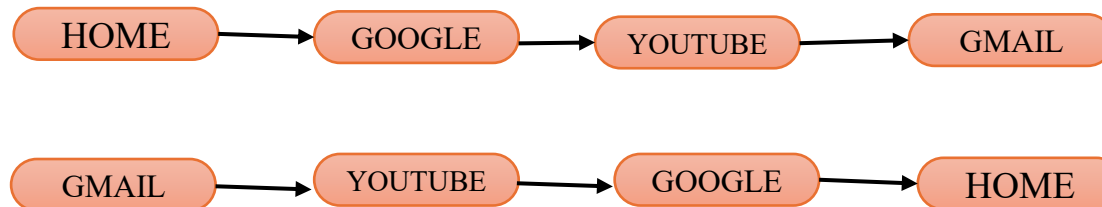
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INTRODUCTION

What is a Stack?

- Stack is a linear data structure.
- It follows the LIFO (Last In, First Out) principle.
- The last visited webpage is the first one returned when the Back button is pressed.
- Operations used:
 - ✓ Push – Store newly visited webpage.
 - ✓ Pop – Remove current webpage while going back.
 - ✓ Peek – Display current webpage.





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Architecture / Algorithm

Working Process

- User opens a webpage.
- Browser pushes the page into the stack.
- Every new page is pushed on top.
- When Back button is clicked, the top page is popped.
- Browser displays the previous page



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Advantages & Performance

Stack is Suitable

- Fast insertion and deletion.
- Simple implementation.
- Efficient memory usage.
- Provides quick Back navigation.
- Widely used in Chrome, Firefox, Edge, and other browsers



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CONCLUSION

- Stack is the most suitable ADT for browser Back navigation.
- LIFO perfectly matches browser history behavior.
- Push and Pop operations take **constant time ($O(1)$)**, ensuring fast performance.
- Stack improves browsing efficiency and provides a smooth user experience.



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THANK YOU